Dataset	$\#F_{total}$	min(|Var(F)|)	max(|Var(F)|)	min(|F|)	max(|F|)
Global	4550	0	8286433	0	7689680
↪ Lagniez [?]	1948	5	229100	10	399794
↪ Soos [?]	1823	2	8286433	1	7689680
↪ Plazar [?]	501	14	486193	31	2598178
↪ Sundermann [?]	278	0	31012	0	350120

Table 1: Dataset summary. The first column indicates the dataset, and the # F column indicates how many satisfiable formulae the dataset contains. The following columns indicate the minimum and maximum number of variables (resp. clauses) in the dataset.

Dataset	$\#F_{total}$	# D4	# ¬D4	# CapKC ∧ ¬D4	# DivKC ∧ ¬CapKC ∧ ¬D4
Global	4550	185	670	80	55
↪ Lagniez [?]	1948	53	213	40	23
↪ Soos [?]	1823	100	393	40	28
↪ Plazar [?]	501	31	61	0	4
↪ Sundermann [?]	278	1	3	0	0

Table 2: Experimental results regarding the scalability of DivKC and CapKC. Column $\#F_{total}$ indicates the total number of formulae in each dataset. The next column shows the number of formulae compiled only by D4 [?] but not by DivKC or CapKC. Column # ¬D4 shows the number of formulae not compiled by D4. Column # CK ∧ ¬D4 shows the number of formulae that are compiled by CapKC but not by D4. The last column indicates the number of formulae that were only compiled by DivKC, but not by D4 or CapKC.

Dataset	$\#F_{total}$	# ¬D4	# ¬D4 ∧ CapKC	# ¬D4 ∧ DivKC	# CapKC ∧ ¬DivKC ∧ ¬D4
Global	4550	670	80	114	21
↪ Lagniez [?]	1948	213	40	52	11
↪ Soos [?]	1823	393	40	58	10
↪ Plazar [?]	501	61	0	4	0
↪ Sundermann [?]	278	3	0	0	0

Table 3: Experimental results regarding the scalability of DivKC and CapKC.

Dataset	# F	$Y_l \leq \|R_F\|$	$Y_h \geq \|R_F\|$	Coverage	$\|R_{G_P}\| \leq \|R_F\| \leq \|R_{G_U}\|$
Global	3670	0.969	0.857	0.826	1.000
↪ Lagniez [?]	1689	0.969	0.895	0.864	1.000
↪ Soos [?]	1307	0.969	0.906	0.875	1.000
↪ Plazar [?]	403	0.968	0.789	0.757	1.000
↪ Sundermann [?]	271	0.967	0.487	0.454	1.000

Table 4: Experimental results for DivKC-AMC. Column # F indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often $|R_F|$ is within the confidence interval $[Y_l; Y_h]$ and thus measures the accuracy of our method.

Dataset	# F	$Y_l \leq \|R_F\|$	$Y_h \geq \|R_F\|$	Coverage	$\|R_F\| \leq \|R_{G_U}\|$
Global	323	0.994	0.988	0.981	1.000
↪ Lagniez [?]	168	0.994	0.988	0.982	1.000
↪ Soos [?]	149	1.000	0.987	0.987	1.000
↪ Plazar [?]	4	1.000	1.000	1.000	1.000
↪ Sundermann [?]	2	0.500	1.000	0.500	1.000

Table 5: Experimental results for CapKC. Column # F indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often $|R_F|$ is within the confidence interval $[Y_l; Y_h]$ and thus measures the accuracy of our method.

Dataset	# F	$Y_l \geq \lvert R_{G_P} \rvert$	$Y_h \leq \lvert R_{G_U} \rvert$	Both	median (r_c)	max (r_c)
Global	3669	0.807	0.836	0.721	4.17e-9	0.54
↪ Lagniez [?]	1689	0.844	0.869	0.760	1.8e-9	0.54
↪ Soos [?]	1307	0.920	0.893	0.834	1.75e-7	0.0962
↪ Plazar [?]	403	0.697	0.769	0.610	1.64e-13	0.0153
↪ Sundermann [?]	270	0.196	0.448	0.093	1.69e-13	0.281

Table 6: Experimental results comparing the bounds obtained with DivKC-AMC and with Lemma ??. Column # F indicates with how many formulae the statistics have been computed. The 'Both' column indicates how often we have $Y_l \geq |R_{G_P}| \wedge Y_l \leq |R_F|$ and $Y_h \leq |R_{G_U}| \wedge Y_h \geq |R_F|$. The last two columns represent the observed median and maximum values of the ratio $r_c = \frac{min(Y_h, |R_{G_U}|) - max(Y_l, |R_{G_P}|)}{|R_{G_U}| - |R_{G_P}|}$, which was calculated exclusively if $Y_l \leq |R_F| \leq Y_h$. The number of formulae on which the last two columns are computed can easily be obtained by multiplying the # F column with the 'Coverage' column in Table 4.

Dataset	# F	$Y_h \leq \lvert R_{G_U} \rvert$	median (r_c)	max (r_c)
Global	323 - 323	0.985	0.00849	0.0257
↪ Lagniez [?]	168 - 168	0.982	0.00835	0.0257
↪ Soos [?]	149 - 149	0.987	0.00885	0.0257
↪ Plazar [?]	4 - 4	1.000	0.00684	0.0257
↪ Sundermann [?]	2 - 2	1.000	0.00981	0.0155

Table 7: CapKC.

Dataset	# F	$l \leq Y_{\text{ApproxMC}} \leq h$	$l \leq Y_{\text{DivKC-AMC}} \leq h$	$l \leq Y_{\text{CapKC-AMC}} \leq h$
Global	2713	0.973	0.871	0.981
↪ Lagniez [?]	1329	1.000	0.889	0.980
↪ Soos [?]	1177	0.964	0.847	0.979
↪ Plazar [?]	185	0.838	0.892	0.995
↪ Sundermann [?]	22	1.000	0.955	1.000

Table 8: CapKC.

Dataset	# F	# DivKC-AMC	$\log_{10}$(min)	mean	median	$\log_{10}$(max)
Global	3188	2129	-3.7	160.7	1.9	5.1
↪ Lagniez [?]	1599	1160	-3.0	239.1	2.1	4.9
↪ Soos [?]	1349	855	-3.6	6.2	1.8	3.6
↪ Plazar [?]	218	97	-3.7	1.3	0.9	1.2
↪ Sundermann [?]	22	17	-1.0	5507.7	12.6	5.1

Table 9: DivKC vs Approxmc7.

Dataset	# F	# CapKC-AMC	$\log_{10}$(min)	mean	median	$\log_{10}$(max)
Global	2514	1554	-2.3	179.1	1.8	5.0
↪ Lagniez [?]	1205	773	-2.3	287.9	2.0	5.0
↪ Soos [?]	1111	651	-2.2	3.3	1.6	2.8
↪ Plazar [?]	176	111	-1.2	2.1	1.5	1.1
↪ Sundermann [?]	22	19	-0.9	4517.2	11.1	5.0

Table 10: CapKC vs Approxmc7.

Dataset	# F	# CapKC-AMC	$\log_{10}$(min)	mean	median	$\log_{10}$(max)
Global	421	182	-2.2	250.3	0.6	5.0
↪ Lagniez [?]	207	92	-2.2	506.3	0.6	5.0
↪ Soos [?]	194	74	-2.2	2.7	0.5	2.2
↪ Plazar [?]	16	12	-0.9	2.8	2.1	0.7
↪ Sundermann [?]	4	4	0.0	5.6	6.6	0.9

Table 11: CapKC vs Approxmc7.